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SUMMARY

Forward Deployed Engineer / Generative AI Engineer with 6+ years of experience building production RAG systems, agentic AI pipelines, and large-scale data platforms on AWS. Currently at PwC, designing AWS Bedrock-based ingestion and retrieval architectures that process enterprise documents. Hands-on depth across LLM orchestration, vector, hybrid retrieval, prompt engineering, and cost optimization for LLM workloads.

TECHNICAL SKILLS

Generative AI / LLM: AWS Bedrock (Agents, AgentCore, Data Automation, Knowledge Bases), RAG architecture, LangChain, CrewAI, MCP (Model Context Protocol), prompt engineering, prompt caching, context window compaction, embeddings, hierarchical chunking, LLM evaluation

Retrieval & Vector Search: Amazon OpenSearch, FAISS, Redis, hybrid search (BM25 + vector), Reciprocal Rank Fusion (RRF), semantic search, re-ranking

BI & Analytics: Power BI, Databricks SQL, semantic models, natural-language-to-SQL, Looker Studio, Tableau

AWS: S3, Lambda, Textract, Comprehend, Kinesis, Firehose, EMR, DynamoDB, Redshift, CloudWatch, API Gateway

Data Engineering: Kafka, Databricks, Apache Spark, Airflow, ETL/ELT pipelines, Hadoop, stream processing, ServiceNow & SharePoint integrations

Languages & APIs: Python, Node.js, SQL, FastAPI, Flask, Django

Databases: PostgreSQL, MySQL, MSSQL, MongoDB, Amazon RDS, SQL Server

DevOps & Tooling: Docker, Nginx, Git, Pytest, Selenium, Power Automate

PROFESSIONAL EXPERIENCE

PricewaterhouseCoopers (PwC), Bangalore

Senior Associate – Gen AI Engineer *April 2025 – Present*

Project: Norfolk Southern – Enterprise AI Platform

- Designed and implemented a configurable RAG ingestion pipeline using **Kafka, AWS S3, Textract, Bedrock (LLMs and Data Automation), OpenSearch, and Comprehend**, processing 1,000+ multi-MB documents in ~3 hours with zero AWS service throttling.
- Built an **automated ETL sync pipeline** pulling documents from **ServiceNow and SharePoint**, detecting new files, updates, and deletions to keep Bedrock knowledge bases continuously in sync with zero manual intervention.
- Implemented **hierarchical chunking and hybrid search (BM25 + vector with RRF)** for business use cases, improving response accuracy and preventing the pipeline from passing irrelevant or truncated chunks to the LLM.
- Reduced chat response latency by ~75% (from ~90 seconds to ~20 seconds) by building an **in-memory retrieval layer with FAISS and Redis** on top of persistent OpenSearch indexes, serving real-time file-upload chat APIs.
- Lowered LLM inference costs with **context window compaction and prompt caching**, and extended retrieval beyond plain text by integrating **Bedrock AgentCore Code Interpreter** to analyze files containing tables, charts, and images.
- Built a **natural-language chat assistant using MCP** to query semantic models and metrics across **Power BI, Databricks, and SQL Server**, enabling C-suite executives to self-serve data insights with auto-generated charts and plain-language explanations.
- Added **episodic memory** to the **Executive Steering Committee knowledge base and agent**, so committee members can quickly look up progress and insights on initiatives across the enterprise.

Additional Engagements: T-Mobile, AbbVie

- Designed a **Databricks ETL pipeline** for T-Mobile performing multi-system data reconciliation and synchronization across 3 operational data sources.
- Developed automated data validation and quality checks in Databricks to detect and alert on discrepancies between T-Mobile's key operational systems.
- Led a **LangChain-based proof of concept** for AbbVie covering entity extraction and document quality validation, improving accuracy of document processing workflows.

Digimaxx AI Solutions Pvt Ltd, Bangalore

AI Technical Lead *June 2024 – March 2025*

- Developed a **multi-agent system with Python and LangChain** that automated competitor price search, collection, and reporting across 10+ e-commerce platforms, replacing hours of daily manual research.
- Built a centralized **AI campaign management platform using CrewAI agents** that automated client interactions and reporting, reducing the digital team's manual campaign workload by ~50%.
- Automated marketing asset creation with **Stable Diffusion** models, cutting content production time by 80% and increasing client engagement rates.
- Integrated **Stable Diffusion and LLMs** into campaign workflows to personalize creatives and copy for targeted audience segments.
- Built and optimized **Looker Studio sales dashboards** for a multi-seller aggregator app, giving sellers actionable performance insights.

Straive, Chennai

Team Lead *June 2022 – April 2024*

- Introduced **LangChain, ChatGPT, and Llama 2** for information retrieval across the team's projects, improving search functionality and insight extraction from diverse data sources.
- Rebuilt legacy Python dashboards as streaming pipelines on **AWS Kinesis, EMR, Lambda, and Firehose**, enabling real-time processing with auto-scaling and no manual server monitoring.
- Delivered 5 proofs of concept spanning vision intelligence, website delta detection, federal law data extraction, financial data point extraction, and LLM-based extraction use cases.
- Containerized microservices with **Docker**, improving fault isolation, deployment consistency, and production security.
- Led effort estimation, resource planning, and code reviews for a 6-member team across 4+ concurrent client projects, improving delivery predictability.
- Interviewed developer candidates and mentored 4+ interns and junior developers on data processing tools and engineering practices.

Senior Software Developer *April 2021 – June 2022*

- Optimized data pipelines with **Asyncio and multithreading**, increasing job throughput by 120% and scaling processing to 100,000+ jobs per day.
- Cut table detection time in 100-page documents from 20 minutes to 5 minutes using **Joblib multiprocessing and pickling**.
- Built a **no-code web scraping microservice** with Selenium and Scrapy, letting non-developers configure custom data acquisition workflows.
- Developed a document annotation pipeline covering metadata extraction, OCR, translation, and table extraction, improving document processing efficiency.
- Applied **spaCy NLP** to extract entities such as dates and amounts from case summaries and bills, and built reusable cloud connectors (AWS S3, Azure Blob, CloudWatch, RDS).

Tivona Global Technologies, Chennai

Software Developer (including Internship) *November 2019 – March 2021*

- Built 15+ REST APIs with **Django** for an online doctor consultation platform, managing doctor, patient, and hospital data with secure handling.
- Integrated the **Razorpay payment gateway** plus SMS and email microservices, enabling paid consultations and direct patient-doctor communication.
- Developed a monitoring dashboard (PHP, Bootstrap, MySQL) with automated data archival and alerting for 30+ livestock IoT hubs deployed in Russia.
- Automated consultation report and billing PDF generation from HTML templates, cutting report preparation time by ~90%, and automated web application testing with Selenium and Pytest.

CERTIFICATIONS

- AWS Partner: Forward Deployed Engineer – Validated, October 2026.
- AWS Certified Generative AI Developer – Professional, August 2026.
- Azure AI Engineer – Associate, June 2025
- IBM Data Engineering Certification, October 2024.
- Google Data Analytics Professional Certificate, October 2022.
- Prompt Engineering for Open Source LLMs, DeepLearning.AI, October 2023.

EDUCATION

Bachelor of Technology, Biotechnology – SASTRA University, Thanjavur, India *2015 – 2019*